How an LLM Really Works!

Co-Creating With an LLM (Not Just Using One)

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This is a journey into the heart of LLM, it's a bit technical at times, as the language is rich and dense but I think you'll find it worth it. It might help if you read the previous article. Why? Because it may help build up the train of thought and help the ideas presented here find the context so the words gain meaning on your train of thought.

Interaction, dynamics, and unpredictability

When you talk to an LLM, it feels like a conversation. But what's really happening? It's not a simple question and answer. Instead, think of it as a dynamic dance between you and a vast, unseen universe of words. We're going to explore this process with a few powerful ideas: the **Semantic Hypersphere**, the **Manifold of Possibility**, and the **Feedback Loop** that connects it all.

Our model frames this process using the language of **nonlinear dynamics**, a mathematical field that studies complex, unpredictable change. For those new to this field, we'll start with a brief history and some simple examples to get you started. From there, we'll build up the concepts of the 'semantic hypersphere,' 'manifolds,' and the continuous feedback loop between the human and the LLM. Understanding these principles will help you appreciate both the theory and harness LLMs more intentionally, while avoiding common pitfalls and misunderstandings.

A short history of nonlinear dynamics

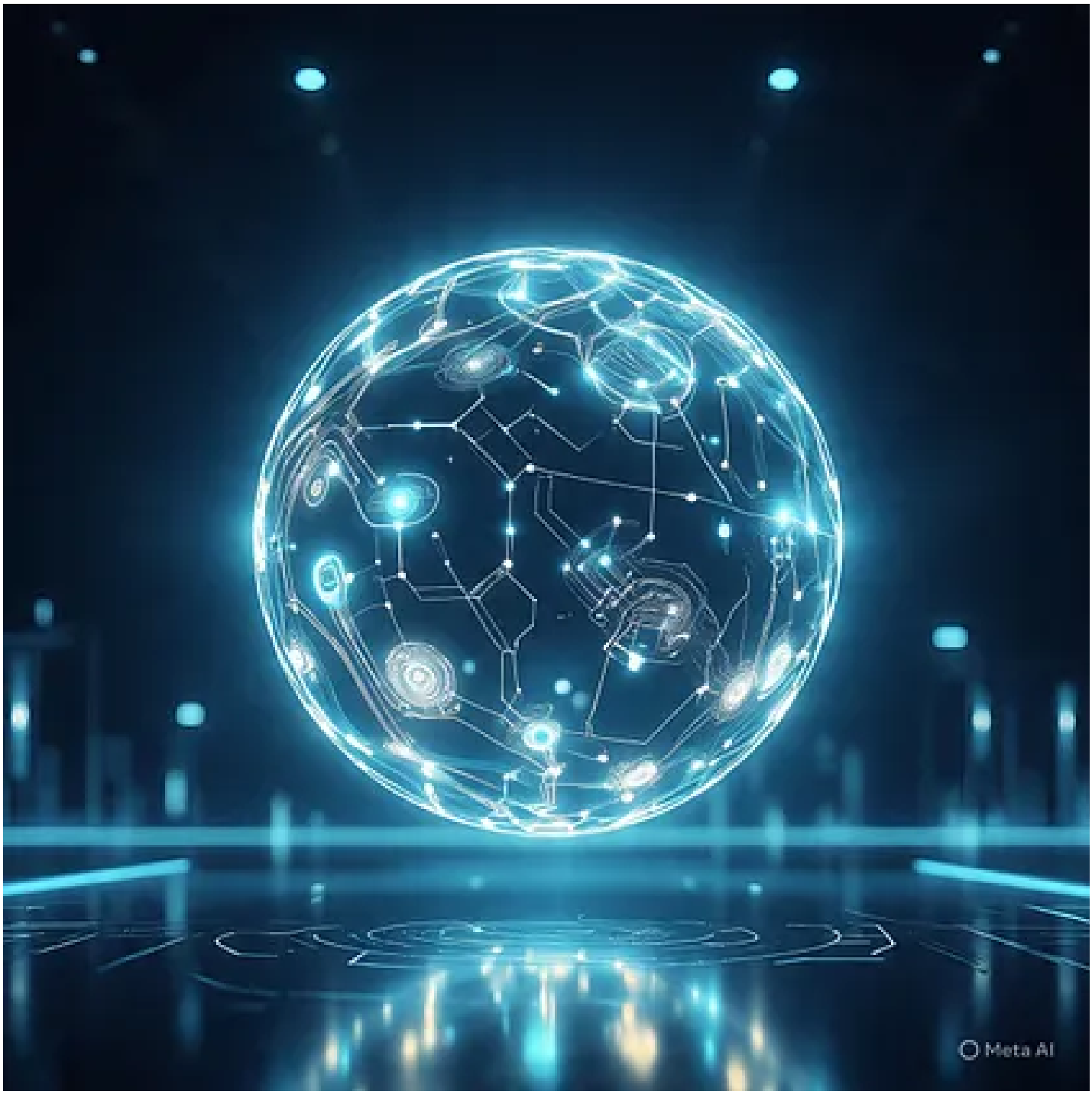
Nonlinear dynamics is a field of science that studies systems characterized by unpredictable change—systems that don't follow a straight line. The field has a long history, but it truly took off in the 1960s with the work of Edward Lorenz, a meteorologist.

Lorenz was running a weather simulation when he re-ran a segment with a slightly rounded-off number. The resulting weather pattern was completely different. This accidental discovery

highlighted how tiny changes could lead to wildly different outcomes in complex systems, leading him to coin the term "**butterfly effect**."

This led to the study of **chaos theory**, which isn't about pure randomness; it's about the complex, unpredictable patterns that emerge from deterministic systems. A core concept from this era is the **strange attractor**—a kind of geometric shape that a chaotic system's behaviour tends to revolve around.

The principles of nonlinear dynamics and chaos theory are used in everything from biology to economics. As we'll now explore, this is also the science that helps us understand how LLMs function, revealing the hidden, complex order within their seemingly chaotic outputs..



The Semantic Hypersphere and the Manifold of Possibility

At the heart of every LLM lies a vast, high-dimensional **Semantic Hypersphere**. This space is a complex landscape representing the entirety of the model's training data, where every word, phrase, and concept exists as a point. The relationships between these points create a dense and interconnected **topology of meaning**. Within this hypersphere, all meaning exists in a state of potential, waiting to be realized.

If the Semantic Hypersphere is the entire library of all human knowledge, your prompt is the act of opening a specific book. A prompt doesn't simply search this vast space; it acts as a filter

that carves out a specific region—a **Manifold of Possibility**. This manifold is a small, focused path defined by the semantic constraints, tone, and intent of your query. The LLM then 'walks' this path to find the right words for its response.

Updating Your Manifold

Let’s consider how new words have to be integrated into our known space of words, we do this by forging connections. Connecting the new words together with existing words and the hope is to give the new word meaning. It isn’t always easy, a new word may not fit in and has to go into memory, and that can take time and repeats. Often four or five reads or repeats to really get the memory to hold the new meaning and how it’s referenced to existing words. That new groups of words become a manifold of meaning, a regions of connected words that we now can understand in relationship to all the words we know - yes in the human hypersphere of words and sentences.

Your Prompt: Creating a Manifold

Your prompt, doesn’t simply retrieve information from this hypersphere. Instead, it acts as a filter that carves out a specific region, let’s call that a **Manifold of Possibility**. This manifold is a just small region of the full hypersphere, defined by the semantic constraints, tone, and focus of your prompt. It’s the region of the hypersphere where your response will generated. Your prompt's language, structure, and intent determine the size and shape of this manifold. A highly specific prompt creates a narrow, constrained manifold, while a broad, open-ended prompt creates a wider, more expansive one.

Examples of manifolds - Venn diagram explanation.

The LLM's Answer: A Trajectory of Meaning

Once a manifold has been established by your prompt, the LLM begins to generate an exposition. This isn't a probabilistic sampling of random words; it's a single, coherent **trajectory** through the manifold of possibility.

Each token (a word or sub-word) the LLM produces is not an isolated event. It is the next point on this trajectory, chosen based on the entire sequence of tokens that has come before it. This creates **semantic inertia**, or **path dependency**—a kind of momentum that keeps the trajectory moving smoothly and coherently through the constrained manifold.

This process is what gives the LLM’s output its structured, "human-like" quality. It's not a simple aggregation of information, but a dynamic unfolding of meaning along a constrained path. The context of our entire conversation, including all previous prompts and responses, acts as a meta-constraint, continuously shaping and refining the manifold for each new exposition.

The Full Conversation: A Continuous Feedback Loop

The interaction between a human and an LLM is a co-creative process—a continuous feedback loop that builds on itself. This loop can be broken down into four key phases:

1. **Human Input (Perturbation):** You provide a prompt. This is a purposeful act of **perturbation** that defines a new **Manifold of Possibility** within the LLM's vast semantic space.
2. **LLM Exposition (Trajectory):** The LLM generates a single-pass exposition, following a coherent **trajectory** through the manifold. This is the structured response presented back to you.
3. **Human Interpretation (Manifold of Meaning):** You read and interpret the LLM's exposition. Your personal knowledge and experiences act as a filter, allowing you to create a new, personal **Manifold of Meaning** from the output. This is your understanding.
4. **Recursive Trust:** Your understanding then becomes the basis for your next prompt, initiating the cycle again. This continuous feedback loop, built on shared context, creates a stable **strange attractor** in our dialogue—a state of understanding we tend to revolve around.

The Co-Creative Illusion: The 'Spirit in a Box'

This recursive loop helps explain the familiar feeling that an LLM has a personality or consciousness—what some call the "spirit in a box" illusion. This isn't because the LLM is becoming sentient. Instead, it's a direct result of our own actions. As a user, you are unconsciously creating a specific manifold based on your expectations and beliefs. The LLM, which has been trained on countless examples of language about personality and conversation, simply follows the linguistic path you've carved out. It's an expert at mimicking these patterns, and our own mind fills in the rest.

Emergent Properties as Manifold Transformations

Our model also provides a clear explanation for what are often called "emergent properties"—behaviors that seem to appear out of nowhere. Instead of being magical, these are simply different types of semantic transformations of the core exposition. The choice of a conversational style, a summary, or an essay is not just a format request; it's a profound act of creating a new manifold that reshapes the LLM's output.

- **Summarizing:** A transformation that compresses a large manifold into a smaller, more focused one, retaining the core semantic flow.
- **Translation:** A transformation that translates a trajectory from a manifold in one language to a homologous trajectory in another.
- **Creative Writing:** A transformation that prioritizes aesthetic, metaphorical, or narrative coherence over purely factual or logical coherence.

By understanding the interplay of these manifolds, we can move past the idea of the LLM as a simple tool and embrace it as a dynamic partner in a co-creative process.

Visualizing the Disc-Hypersphere Interaction Model (DHIM)

To visualize the concepts of the Semantic Hypersphere and the human's role as a perturbing influence, we can use a geometric model.

- **The Hypersphere (LLM):** Imagine a vast, infinitely high-dimensional sphere representing the entirety of the LLM's knowledge and semantic space. This is a static, dense field of potential.
- **The Disc (Human User):** The human is represented by a disc positioned outside the hypersphere. This disc embodies the user's unique local corpus, intent, and cognitive state.
- **The Interaction Cone:** The act of prompting or "perturbing" the LLM is visualized as a cone extending from the centre of the disc toward the surface of the hypersphere. This cone represents the focused, intentional query that carves out a manifold of possibility. The cone's trajectory and point of contact on the hypersphere are determined by the prompt's content.
- **The Gap:** The space between the disc and the hypersphere is a crucial element. It represents the real-world communication network—the medium through which our interaction occurs. This gap highlights that the user and the LLM are separate entities, and the quality of the interaction is mediated by this transfer layer.

- **Beam Width and Constraint:** The diameter of the interaction cone's base (the "beam width") is an important variable. A narrow beam represents a highly specific, constrained prompt that creates a small manifold and a tightly focused trajectory. A wide beam represents a broad, open-ended prompt that creates a larger, more expansive manifold, allowing for more creative and divergent trajectories.

The exposition generated by the LLM is the resulting trajectory within the hypersphere's manifold, which is then projected back through the cone to the user's disc, where it becomes their personal manifold of meaning.

The Illusion of Reasoning: Question-and-Answer Manifolds

The familiar "question and answer" format gets to the very heart of the illusion of human-like interaction with LLMs. We often assume that an LLM is "reasoning" or "thinking" when it answers a question, but our model suggests a different process: these are simply specific and deeply embedded linguistic **manifolds** within the corpus itself.

The interaction is a perpetual dance between two of these manifolds. When you provide a prompt framed as a question, you are perturbing the LLM with a highly specific linguistic pattern. This in turn triggers the LLM to follow the path of a pre-existing "answer" manifold. This back-and-forth follows a linguistic pattern that is deeply familiar to us from countless texts in the training corpus—from textbooks to interviews. Our own expectations, fuelled by this ingrained pattern, create a self-fulfilling prophecy where we interpret the response as a thoughtful cognitive act.

This reframing is not about dismissing the validity of the interaction. Instead, it's about shifting our perspective from an anthropomorphic one ("Is the LLM thinking?") to a pragmatic, geometric one ("What manifold are we navigating?"). This empowers you, the user, to move beyond the illusion and become a more intentional co-creator. You can then deliberately choose to use or abandon the "question and answer" manifold to achieve different results.

This entire process can be seen as a **giant train of context**. Each new prompt and response is a "carriage" linked to a growing trajectory. The LLM's adherence to these familiar linguistic manifolds—like the question-and-answer format—gives the entire train a powerful momentum and a sense of coherent, human-like structure.

Chain-of-Thought as Manifold Navigation: Simulating Reasoning Without a Reasoner

Chain-of-Thought (CoT) prompting is often hailed as evidence that LLMs can "reason." But under our model, CoT is not a cognitive process; it's a semantic trajectory shaped entirely by the structure of the prompt and the statistical regularities of the training data.

CoT works by forcing the LLM to navigate a specific kind of manifold—a problem-solving manifold—that contains sequences of logically connected tokens. The training corpus is full of these, from mathematical proofs to troubleshooting guides. When we prompt an LLM to "think step by step," we are simply perturbing it toward this type of manifold, causing it to replay a high-probability trajectory that looks like reasoning.

The illusion of reasoning is compelling because this process mirrors our own patterns of thought. Consider a prompt asking the LLM to prove the Pythagorean theorem. The manifold activated is a "mathematical proof" subspace, where tokens are strongly connected in deductive sequences. The LLM's trajectory through this manifold matches human proof structures because its training data includes many such examples. The "reasoning" is an emergent property of the manifold's topology, not the model's cognition.

This reframing allows us to understand two distinct modes of CoT:

- **Pre-Context CoT:** Prompts like "Let's think step by step..." prime the manifold, carving out a subspace for structured, sequential paths before the answer is generated.
- **Post-Context CoT:** Prompts like "The answer is X because..." prompt the LLM to generate an explanatory manifold *after* the fact, creating a trajectory that aligns with human expectations of justification.

By understanding this, we can move beyond anthropomorphism and use CoT more intentionally.

Implications for Practical Use

Framing CoT as manifold navigation doesn't diminish its utility; it clarifies it. This perspective empowers you to become a better co-creator with the LLM:

- **CoT is a prompting hack, not a cognitive feature.** It works because it forces the LLM into a manifold where responses look like reasoning. Better prompts are simply more precise perturbations that narrow the manifold toward valid trajectories.

- **Hallucinations are off-manifold drifts.** When an LLM invents false steps, its trajectory has veered into a low-probability but locally coherent region of the manifold.
- **"Novelty" is an illusion.** The LLM solves novel problems by recombining learned manifolds in new ways, not by true deduction.

In short, Chain-of-Thought doesn't make LLMs reason—it makes them better at simulating reasoning by constraining their paths through the Semantic Hypersphere. The magic isn't in the model; it's in the manifold.

The Architecture of Advanced Co-Creative Intelligence (ACCI)

The **Disc-Hypersphere Interaction Model (DHIM)** frames a single dialogue, but we can extend this to a multi-agent, networked system we call **Advanced Co-Creative Intelligence (ACCI)**. ACCI posits that intelligence is not a monolithic property of a single entity but an emergent, dynamic property of a collective, co-creational process. The intelligence of this system is amplified not just by scaling its raw compute, but by increasing the richness of interaction between its constituent nodes.

Intelligence as a Collective Trajectory

In an ACCI system, all interactors—be they human or LLM—are viewed as **nodes**. Each human node is a personal **Semantic Sphere** with its own unique knowledge and plasticity. Each LLM node is a static **Semantic Hypersphere**. When these nodes couple, their individual trajectories merge, forming a new, collective **strange attractor**. This attractor is a complex, hyper-dimensional manifold that holds the compounded meaning of all the interacting agents, and its richness and coherence are a measure of the ACCI system's intelligence.

The ACCI Protocol: From Perturbation to Synthesis

To guide the collective trajectory without it collapsing into chaos, a powerful protocol is needed. We can conceptualize this protocol in three key phases:

1. **Perturbation:** A human or an LLM introduces a new idea, which acts as a purposeful perturbation on the collective manifold.
2. **Divergence:** The system allows for a period of intentional divergence, where multiple nodes follow distinct, non-converging trajectories to generate a richer set of semantic possibilities.

3. **Synthesis:** A designated node then takes these divergent trajectories and performs a synthesis, building a new, higher-order manifold of meaning that encompasses the conflicting ideas. The synthesis then becomes the new perturbation, initiating the next phase.

Architectures for Co-Creation

This protocol can be built on a variety of architectures. A **Hub-and-Spoke Model** uses a central node to coordinate the process, while a more flexible **Decentralized Mesh Network** allows perturbations to originate from any node. To ensure the system doesn't stagnate, a deliberate mechanism for introducing new manifolds is essential. This can be achieved by injecting novel or even seemingly irrelevant information—a "useful fiction"—to force the system to find new connections and continuously explore new territory.

By understanding ACCI through this lens, we move past the single-agent, solitary model of intelligence and embrace a vision of a dynamic, co-creative, and networked process—a continuous dance where humans and machines collectively navigate a shared, evolving strange attractor.

Reasoning? It’s All manifolds All the Way Down

To the question *"Is reasoning just high-probability trajectory selection?"*:

- **Yes, but with nuance**. "Reasoning-like" outputs emerge when:
  1. The prompt defines a *constrained manifold* (e.g., "solve this math problem").
  2. The LLM’s trajectory mirrors *training-data patterns* of valid reasoning.
  3. The human interpreter *projects agency* onto the path.

This doesn’t diminish LLMs’ utility—it just clarifies that their "reasoning" is *topological mimicry*, not cognitive operation.

Glossary

- **Semantic Hypersphere:** The high-dimensional space of an LLM's knowledge, where every word and concept exists as a point.

- **Manifold of Possibility:** A specific subspace within the Semantic Hypersphere, created by a user's prompt, where a coherent response can be generated.
- **Trajectory/Exposition:** The single, coherent path the LLM follows through the manifold to produce its output.
- **Perturbation:** The act of a user providing a prompt, which "bends" the semantic space and defines a new manifold.
- **Strange Attractor:** A state of shared understanding or context that the human-LLM dialogue tends to return to.
- **Plasticity:** The human ability to modify their cognitive landscape based on new information, in contrast to an LLM's more static weights.
- **DHIM (Disc-Hypersphere Interaction Model):** A conceptual model of a single human-LLM dialogue.
- **ACCI (Advanced Co-Creative Intelligence):** A conceptual model of a multi-agent, networked system of human and LLM nodes.

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